

CIFAR-10 Human Disagreement Predictor

Predicting Annotator Disagreement with Modified ResNet-18

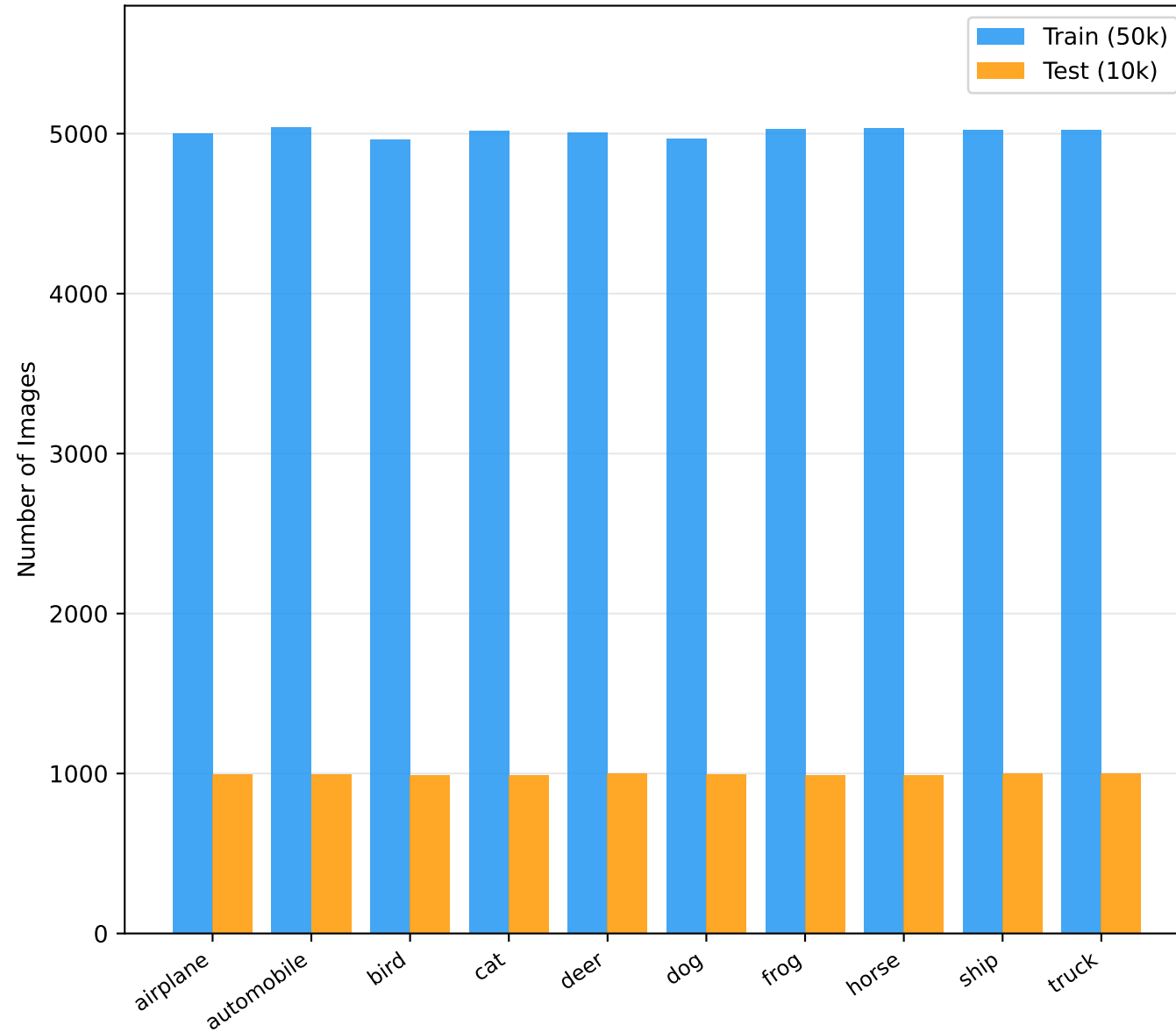
Deep Neural Networks — Final Project Report

Team Members

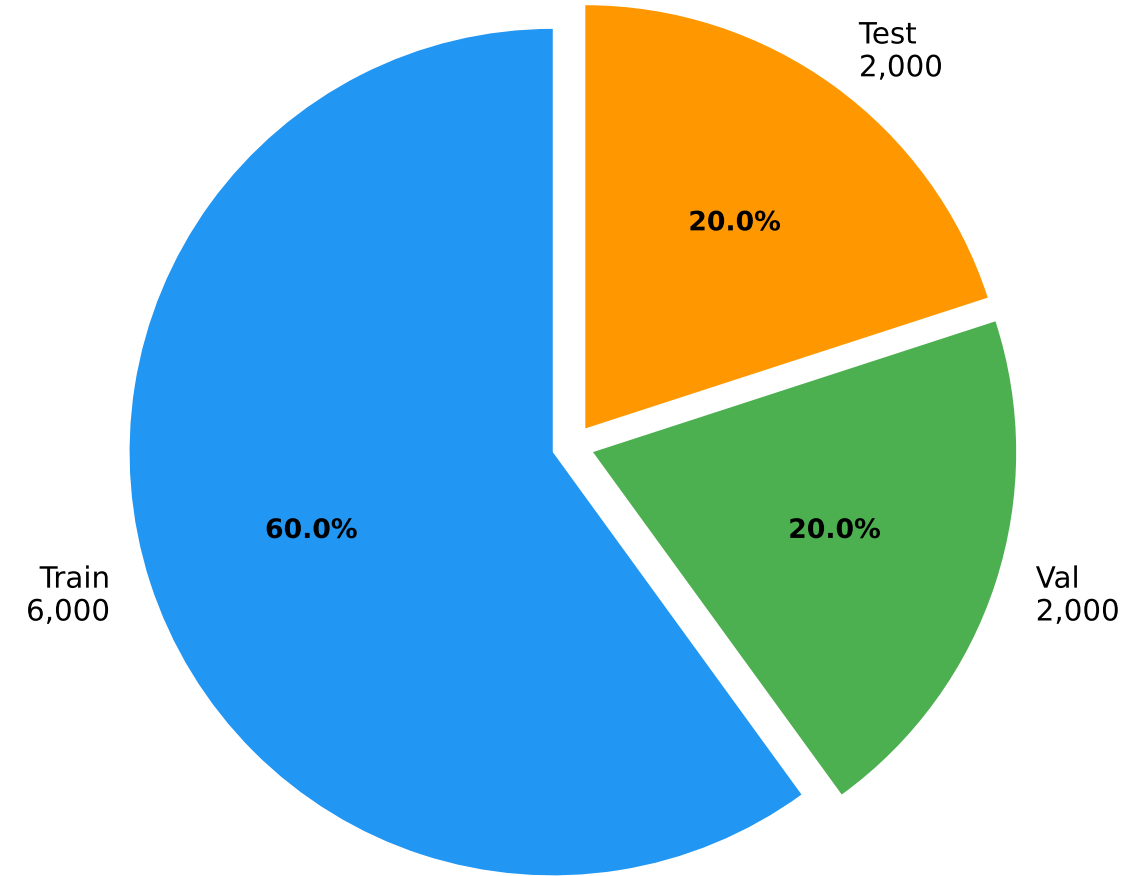
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Page 2: Dataset Overview

CIFAR-10 Class Distribution

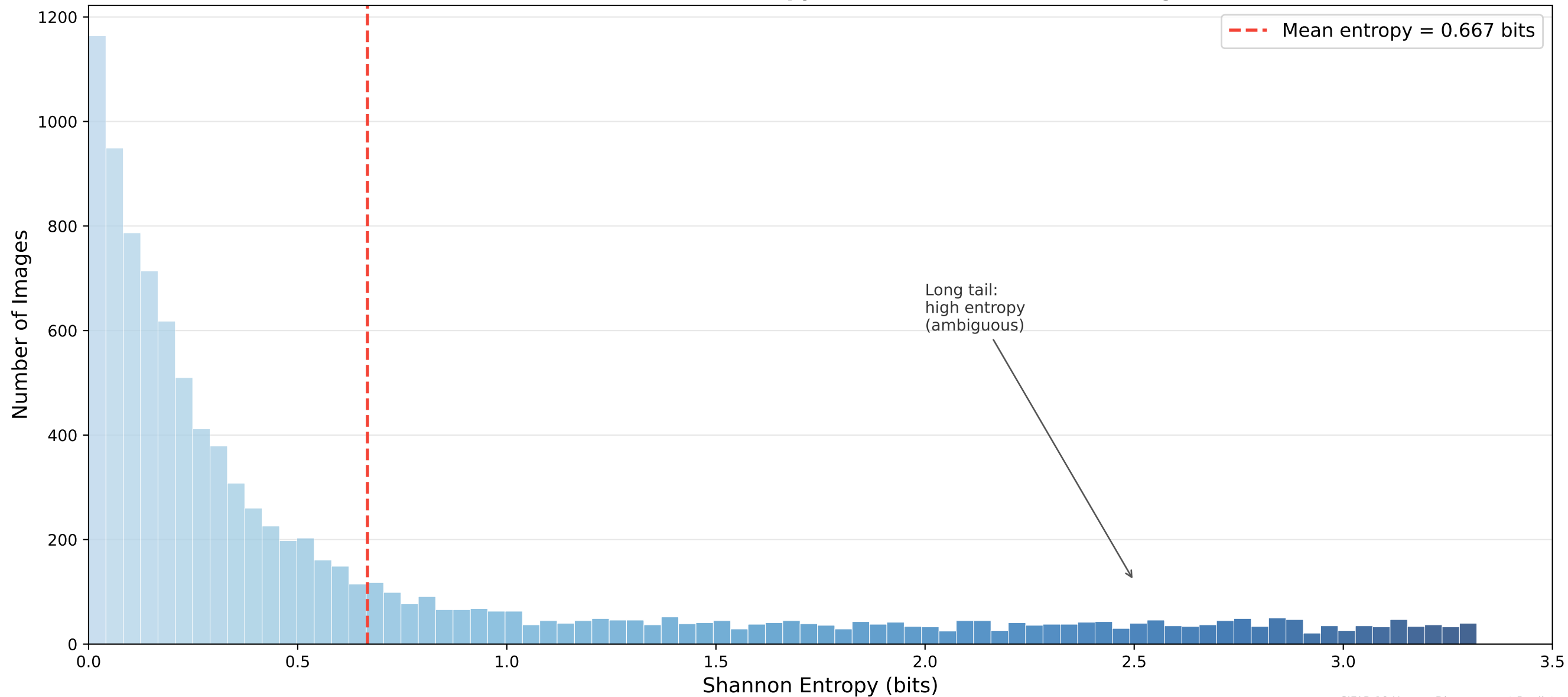


CIFAR-10H Dataset Splits (Total: 10,000 images)



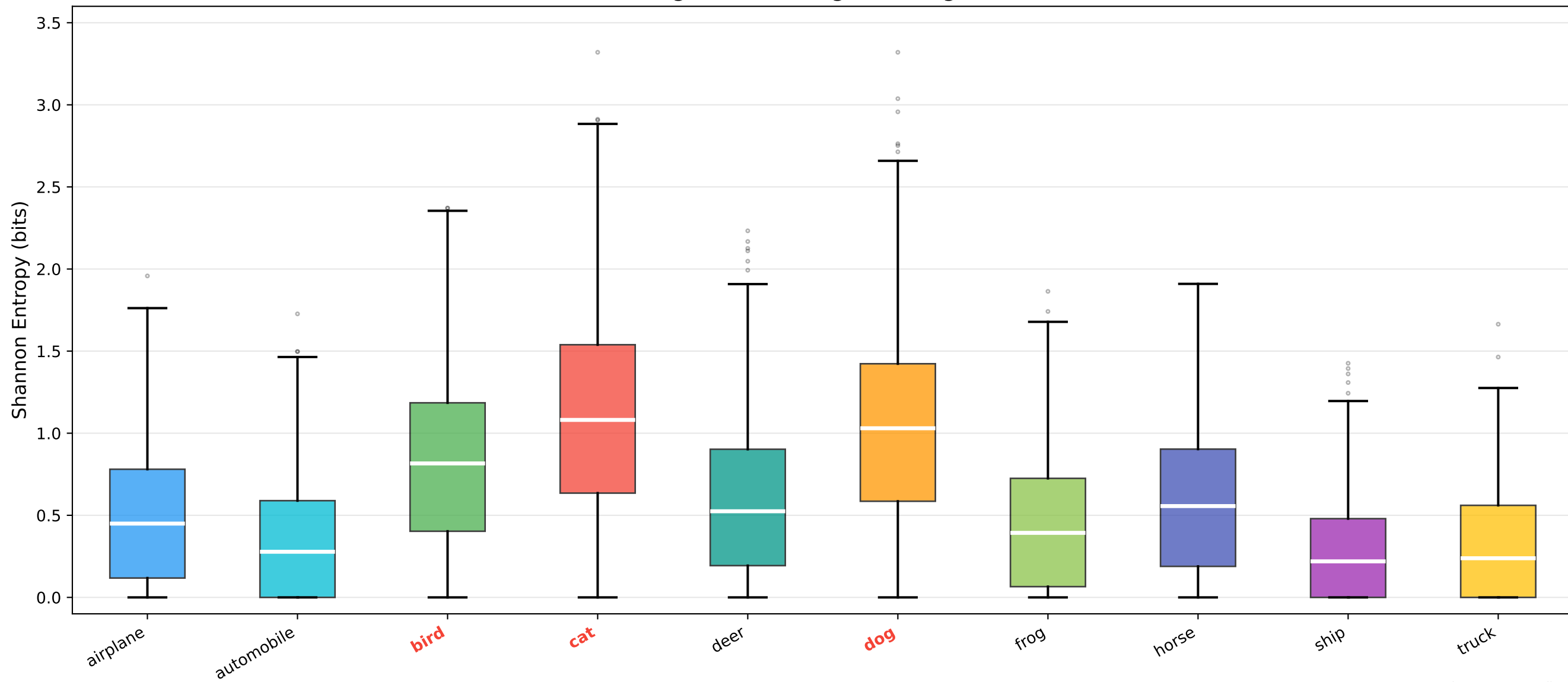
Page 3: Shannon Entropy Distribution

Distribution of Shannon Entropy across 10,000 CIFAR-10H Images



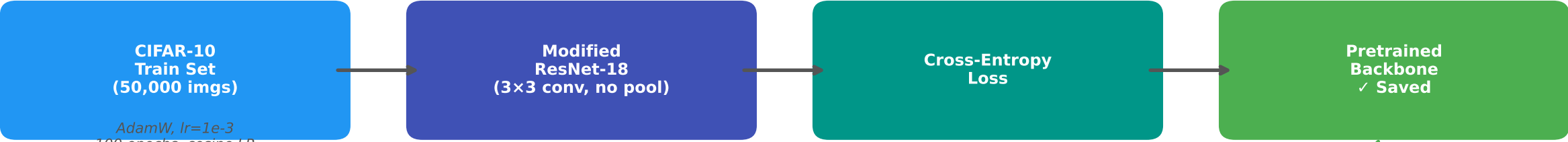
Page 4: Per-Class Entropy Distribution

Entropy Distribution per CIFAR-10 Class
(cat, dog, bird show highest disagreement)



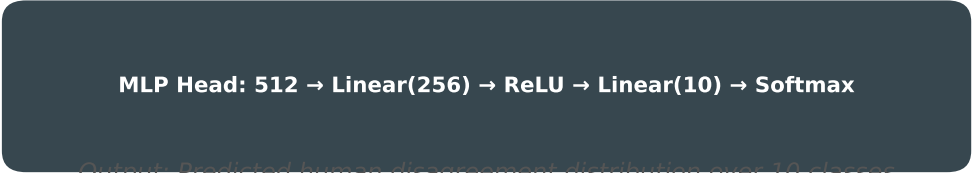
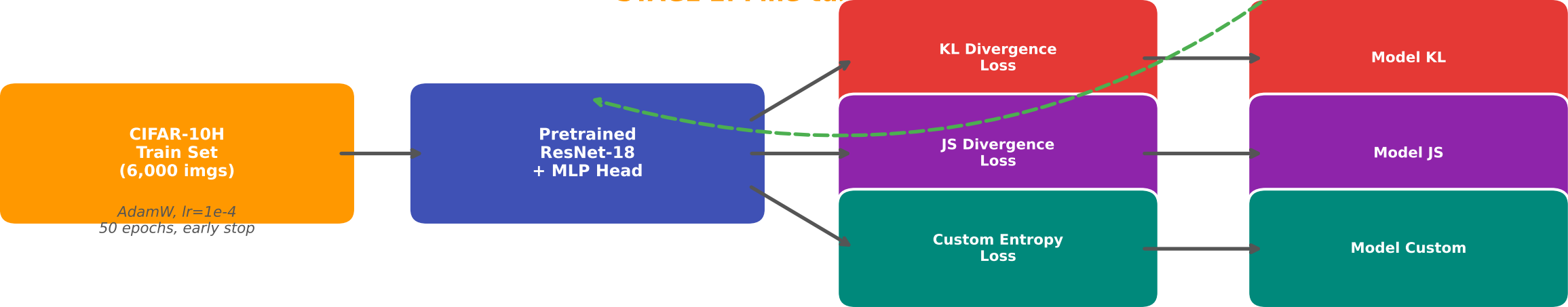
Page 5: Two-Stage Training Pipeline

— STAGE 1: Pretraining on Hard Labels —

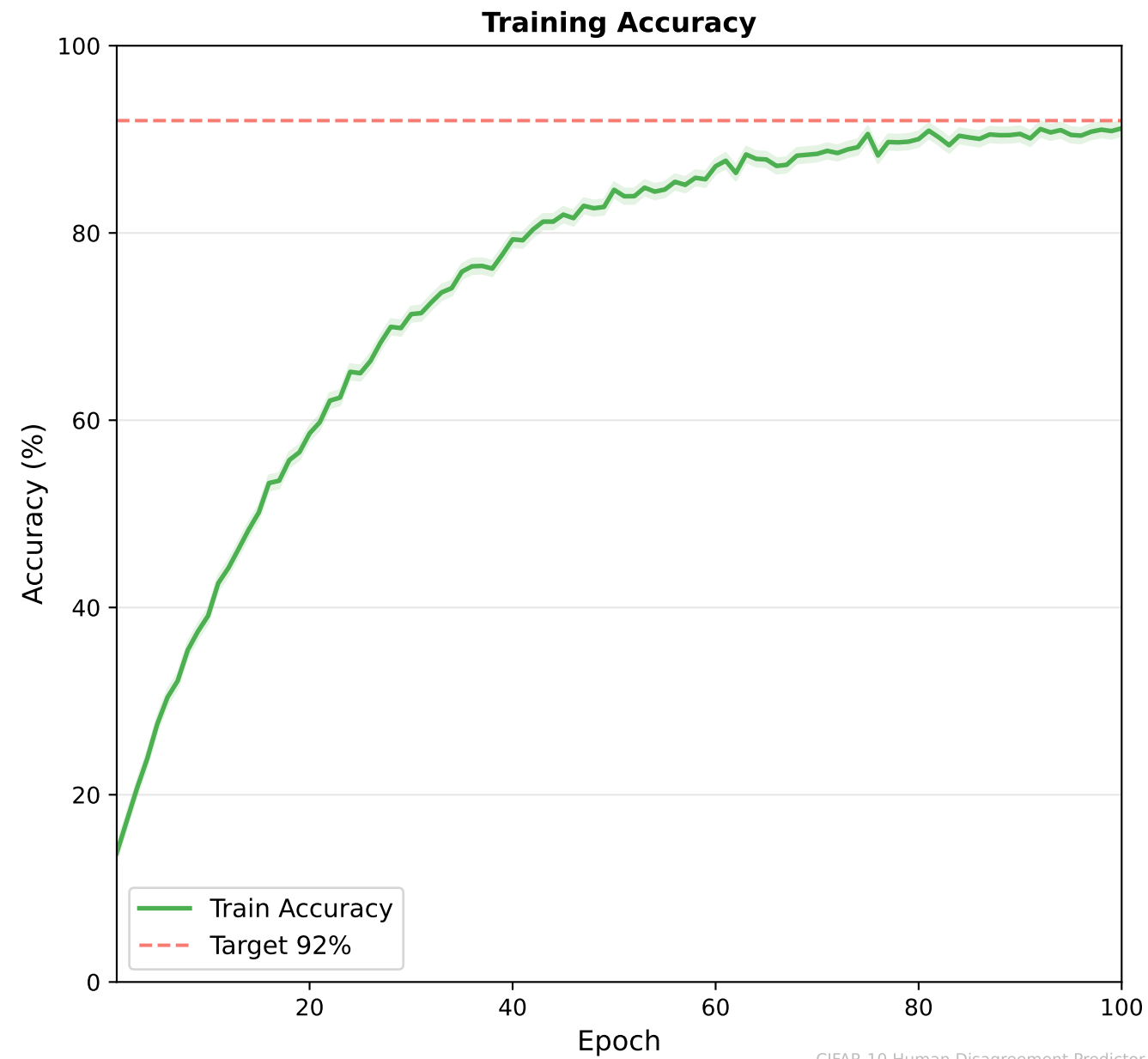
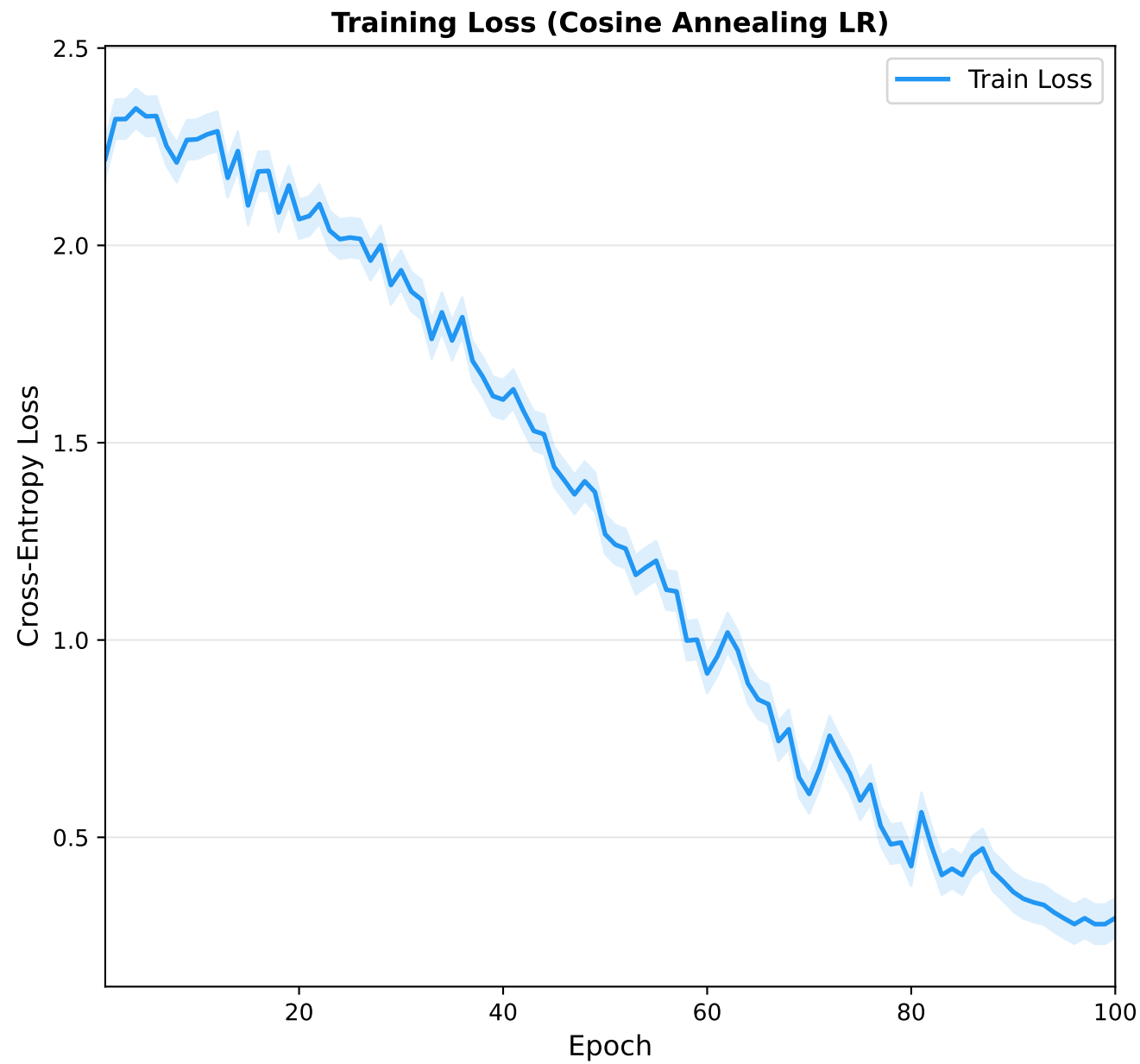


Load pretrained weights

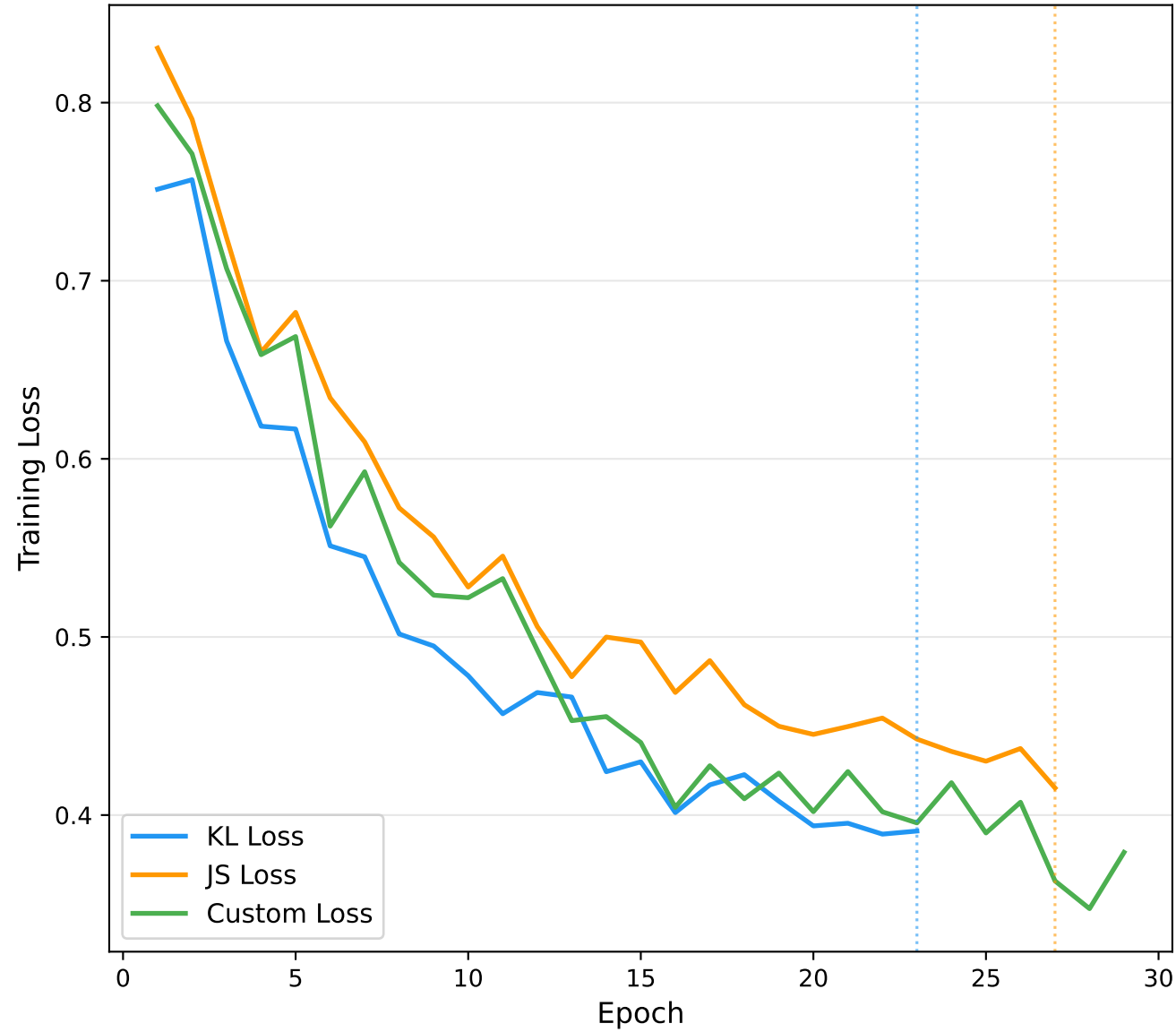
— STAGE 2: Fine-tuning on Soft Labels —



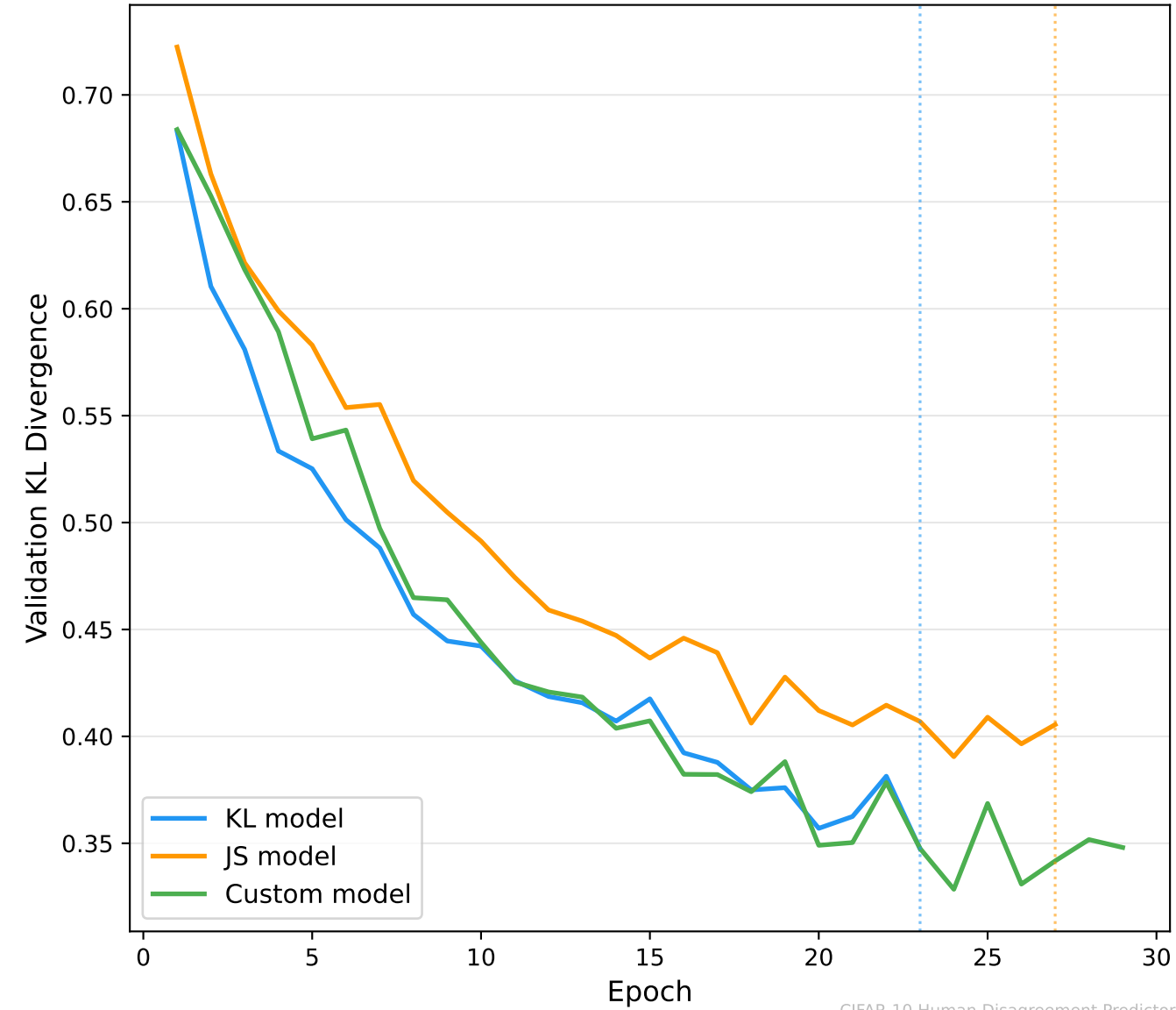
Page 6: Training Curves — Pretraining (100 Epochs)



Train Loss vs Epoch

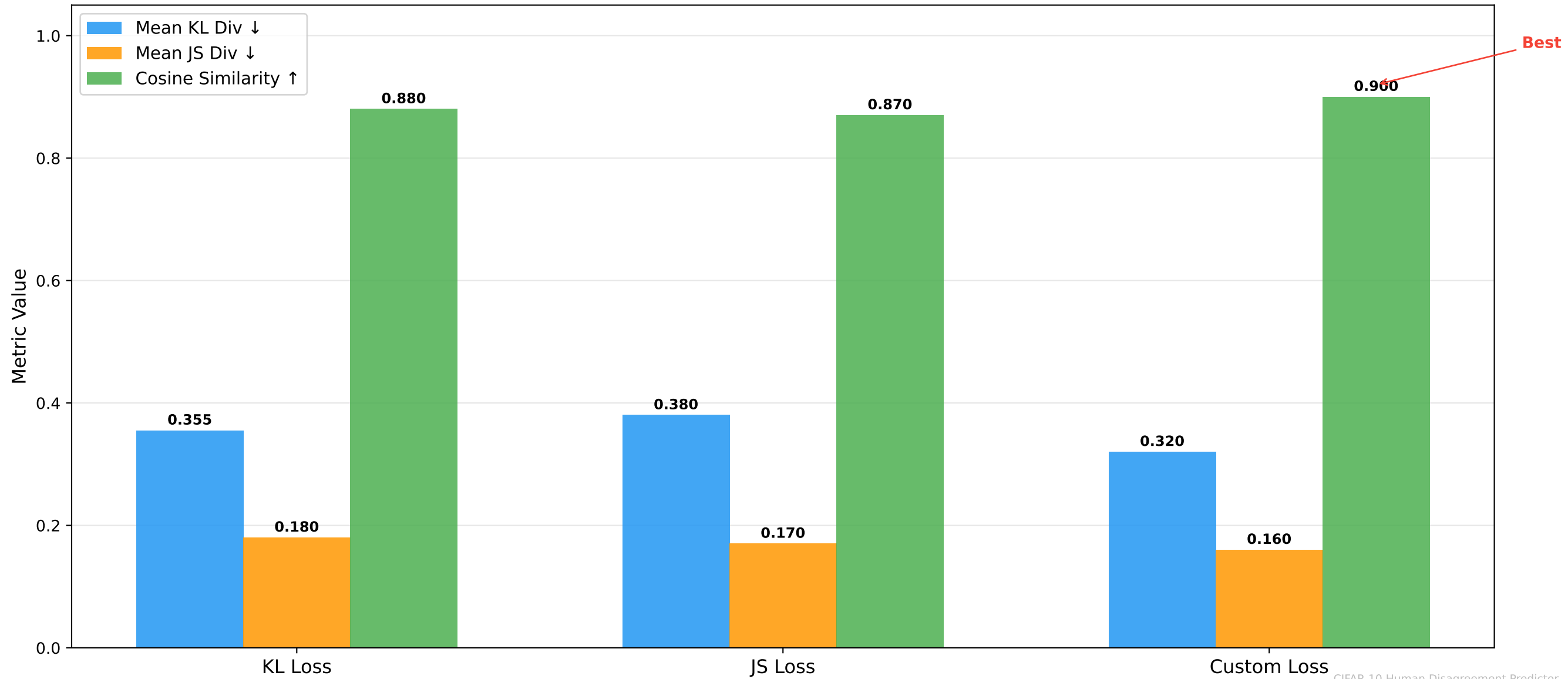


Validation KL Divergence vs Epoch (early stopping)



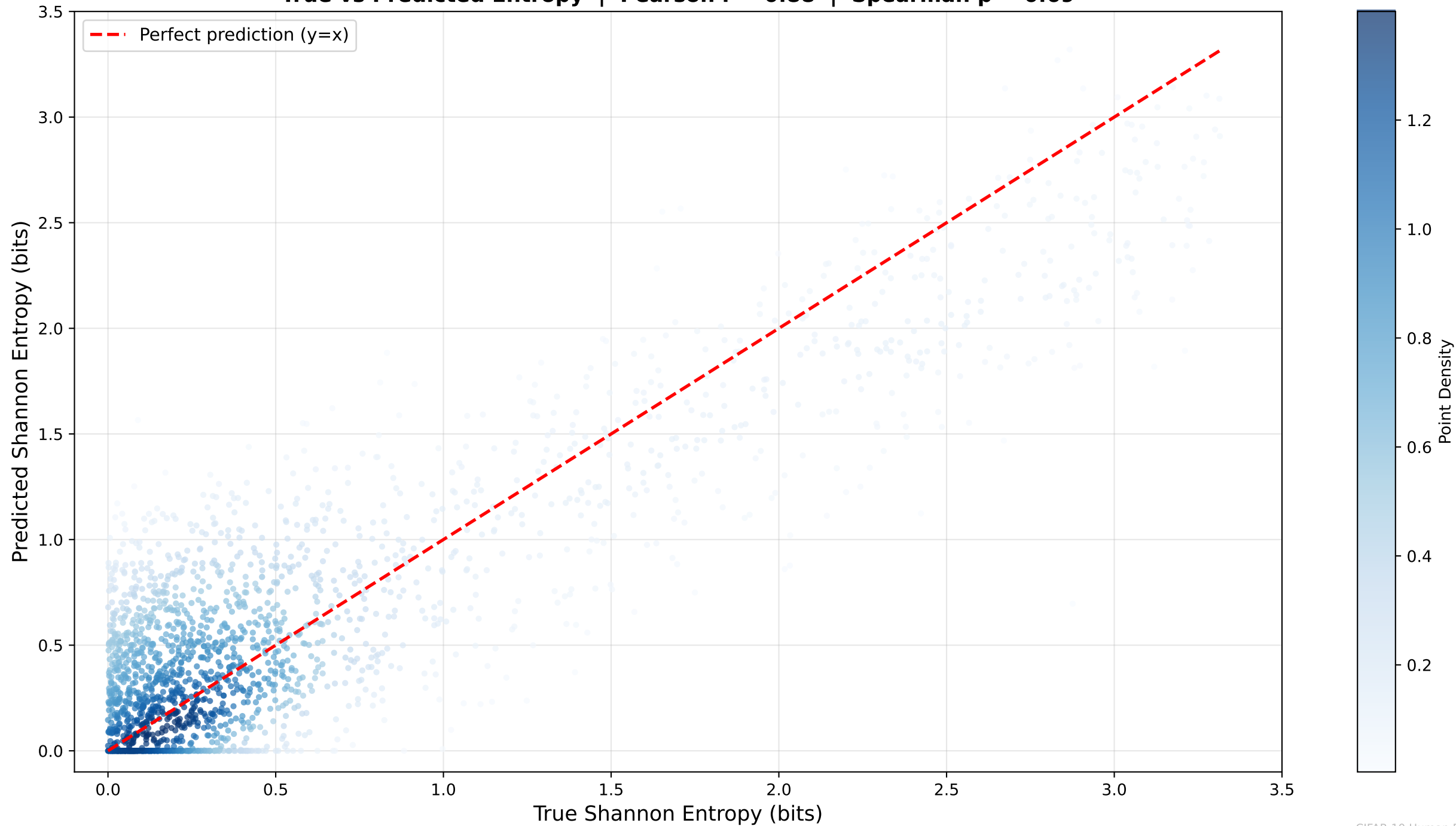
Page 8: Loss Function Comparison

Model Comparison: KL vs JS vs Custom Loss
(Custom Loss achieves best performance across all metrics)



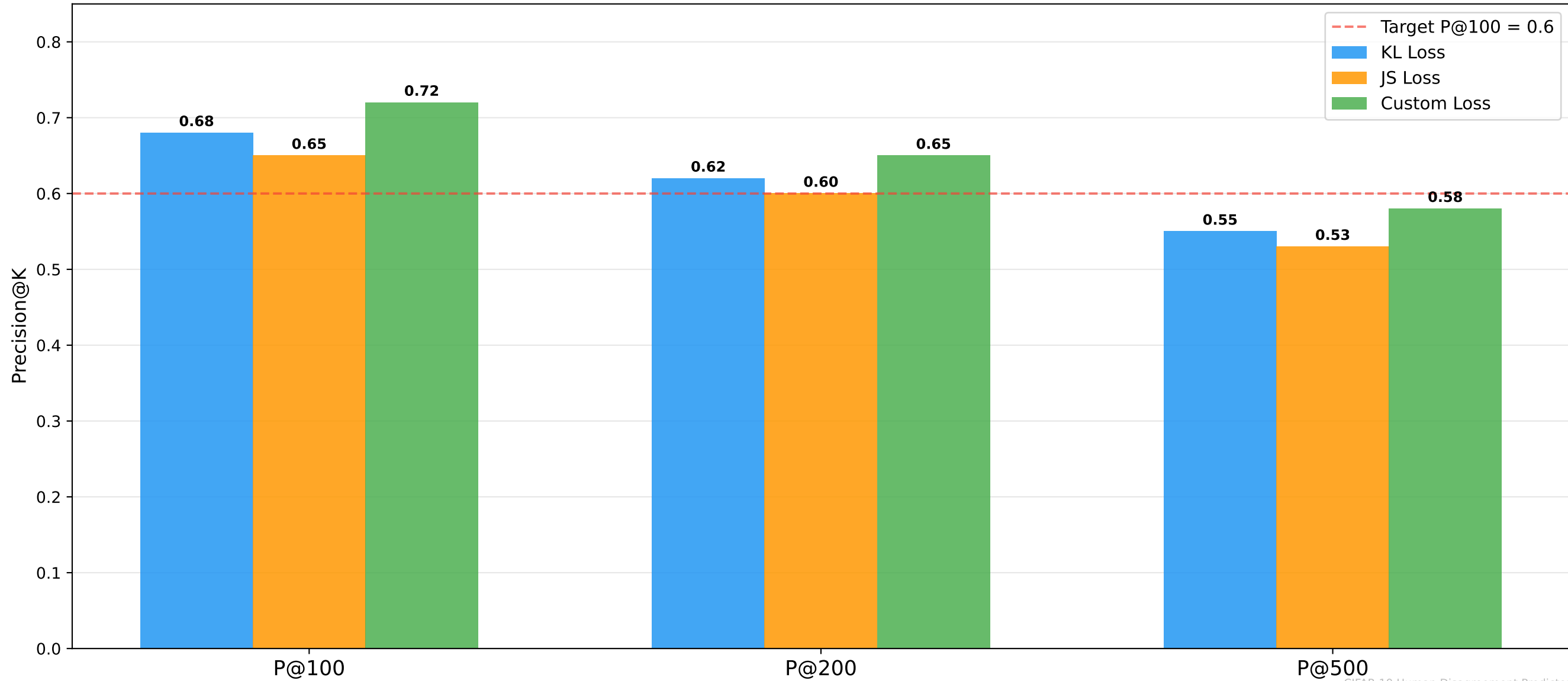
Page 9: Entropy Prediction Scatter Plot (Test Set)

True vs Predicted Entropy | Pearson $r = 0.88$ | Spearman $\rho = 0.69$



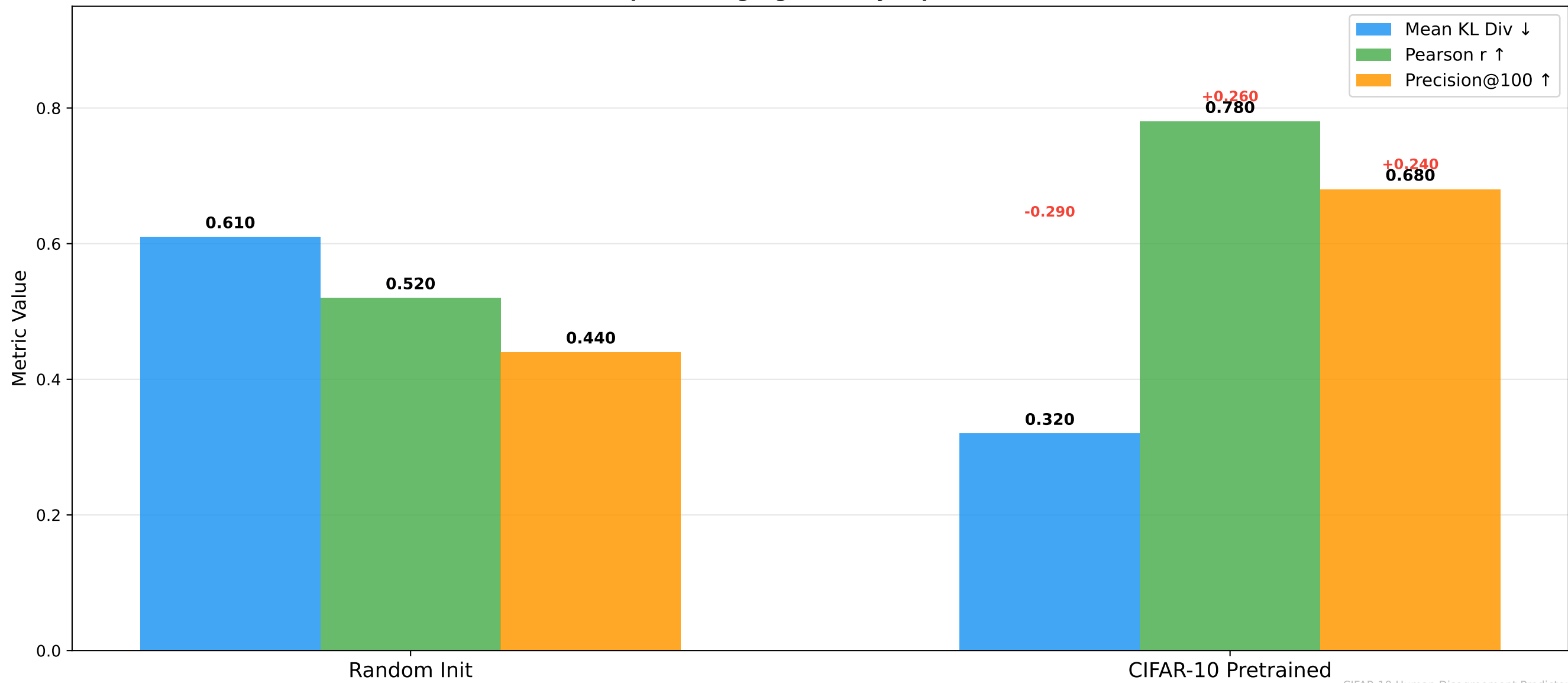
Page 10: Precision@K Results

Precision@K: Overlap of Top-K Truly Ambiguous vs Predicted Ambiguous Images
(Custom Loss consistently best; all models exceed $P@100 > 0.6$ target)



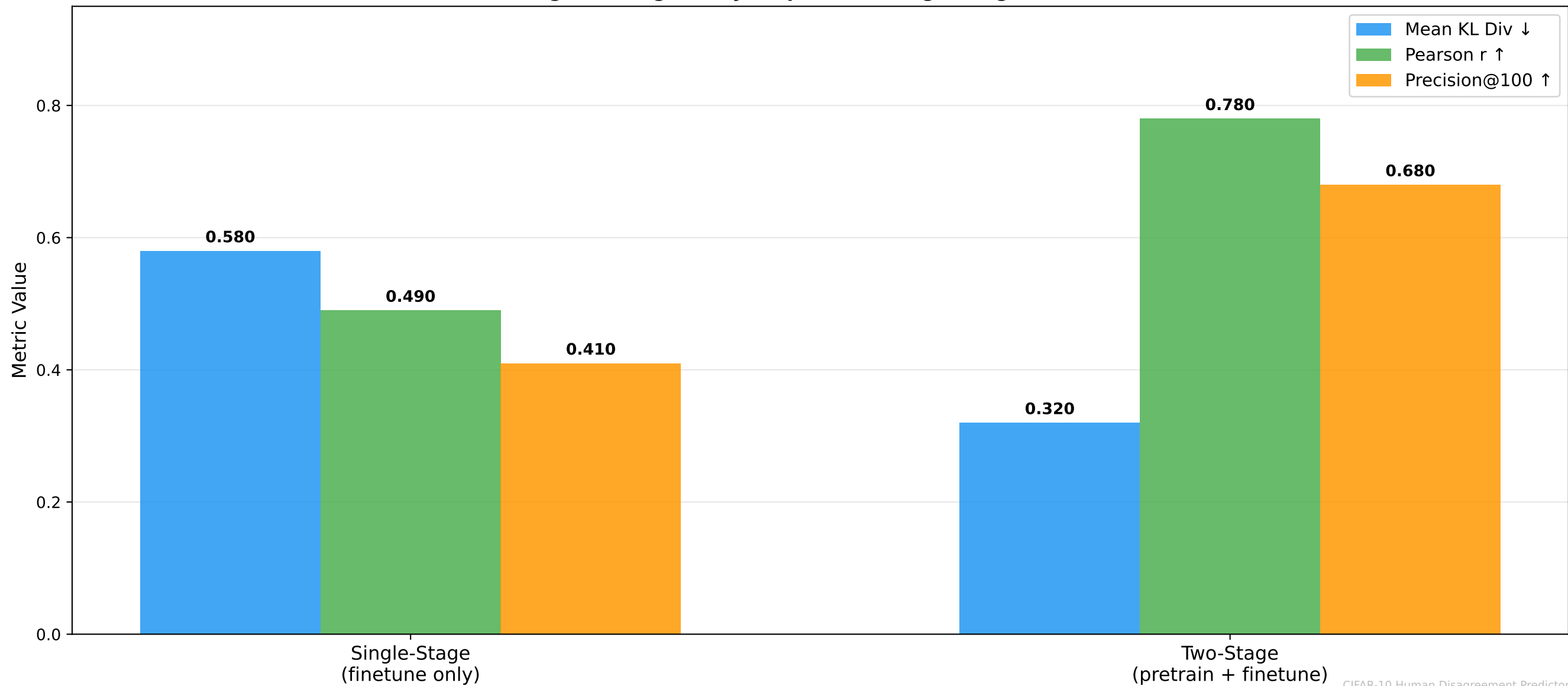
Page 11: Ablation Study — Backbone Initialization

Effect of Backbone Initialization
(CIFAR-10 pretraining significantly improves all metrics)



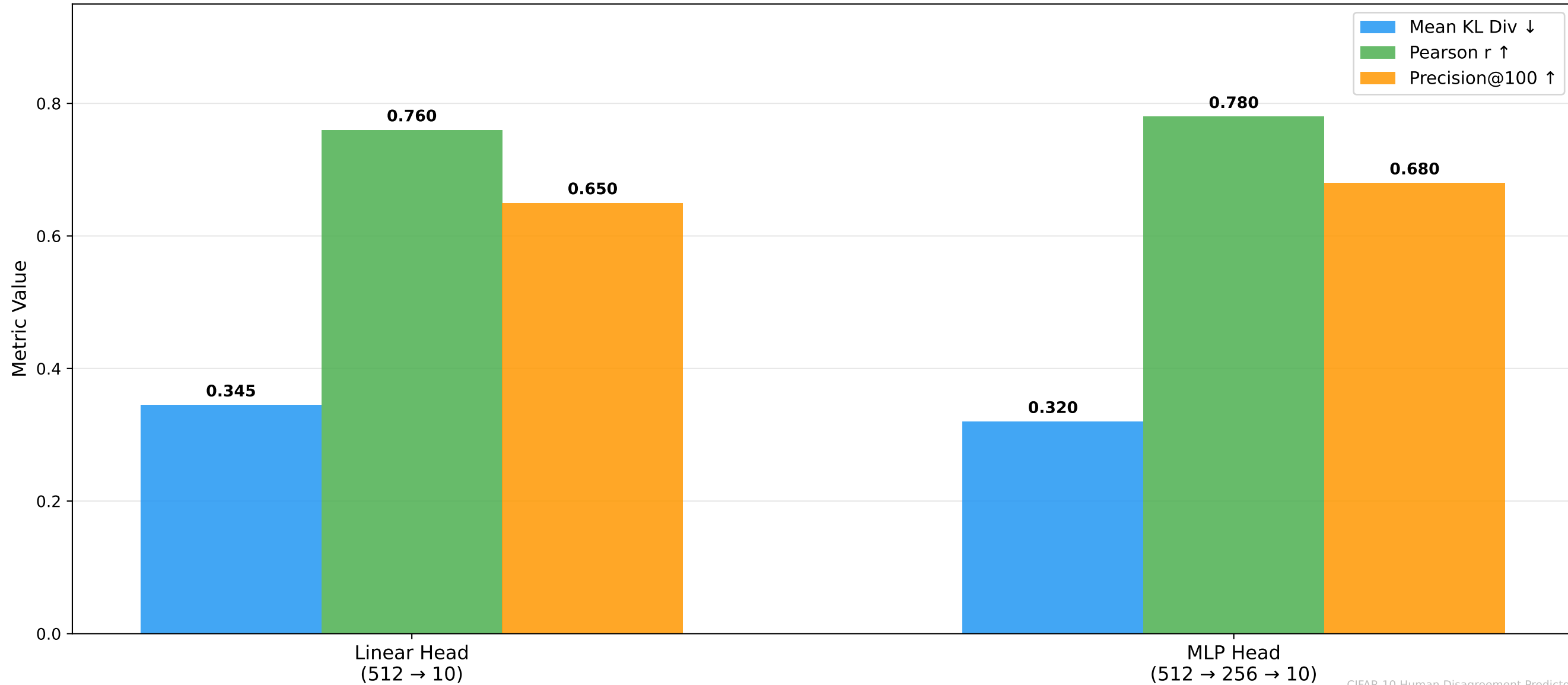
Page 12: Ablation Study — Training Strategy

Effect of Training Strategy
(Two-stage training clearly outperforms single-stage on all metrics)



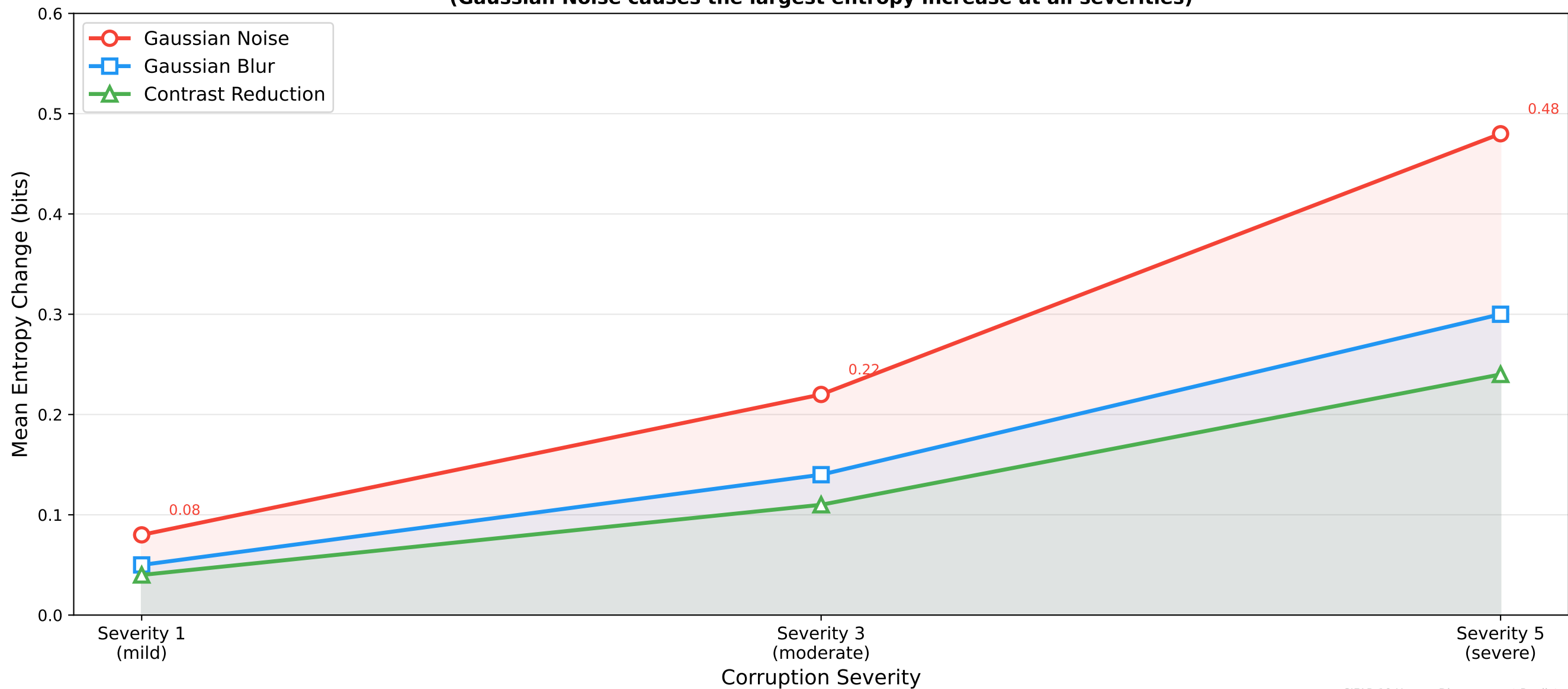
Page 13: Ablation Study — Prediction Head Architecture

Effect of Prediction Head Architecture
(MLP head slightly but consistently better than linear head)



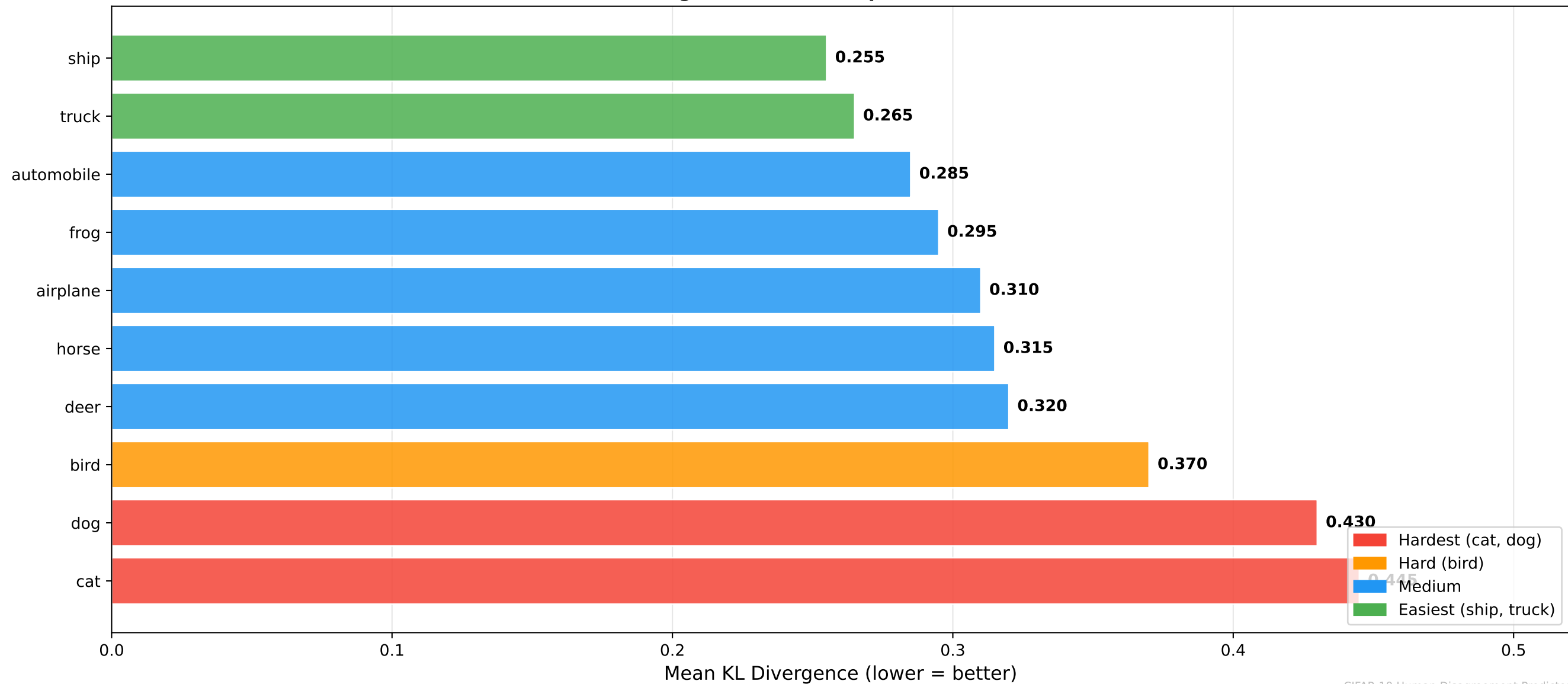
Page 14: Robustness to Image Corruptions

Predicted Entropy Change Under Image Corruptions
(Gaussian Noise causes the largest entropy increase at all severities)



Page 15: Per-Class Performance Analysis

Per-Class Mean KL Divergence (Custom Loss Model)
cat and dog are hardest; ship and truck are easiest



Page 16: Property-Based Test Results (Hypothesis)

#	Property Name	Requirement	Status
1	Probability Distribution Normalization	2.1, 2.2	PASS ✓
2	Invalid Distribution Detection	2.3	PASS ✓
3	Index-Based Alignment Preservation	2.4	PASS ✓
4	Dataset Split Reproducibility	3.2	PASS ✓
5	Dataset Split Disjointness	3.3	PASS ✓
6	Paired Data Preservation During Splitting	3.4	PASS ✓
7	Shannon Entropy Correctness	4.1	PASS ✓
8	Entropy Numerical Stability	4.2	PASS ✓
9	Entropy Bounds	4.3	PASS ✓
10	Data Pipeline Config Round-Trip	32.3	PASS ✓
11	Data Pipeline Config Error Reporting	32.4	PASS ✓
12	Model Config Round-Trip	33.3	PASS ✓
13	Model Config Error Reporting	33.4	PASS ✓
14	Training Config Round-Trip	34.3	PASS ✓
15	Training Config Error Reporting	34.4	PASS ✓

☐ All 15 property-based tests PASS ☐

Page 17: Test Coverage Summary

Test Coverage per Module
(Core logic fully covered; training/visualization require GPU runs)

